

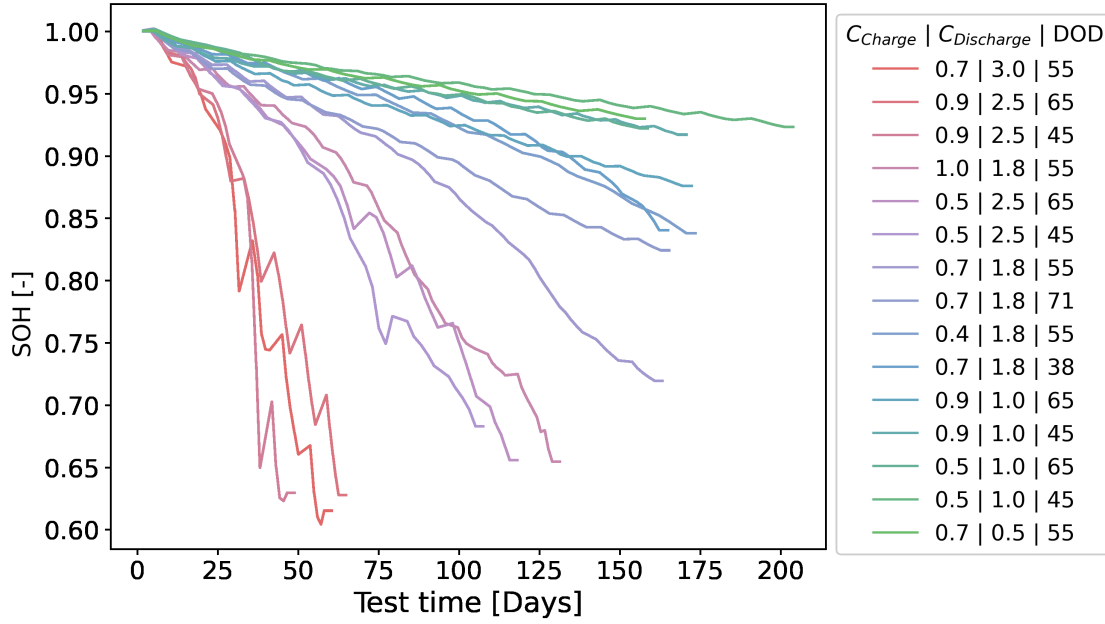


Figure 4.4: Measured SOH progression across the 15 LFP cells over the complete ageing campaign. Colors distinguish the fixed training, validation, and independent holdout test sets. The legend reports the charge C-rate, discharge C-rate, and depth of discharge assigned to each trajectory. Every SOH trajectory is constructed from discrete capacity check-ups and interpolated between the measured anchors.

The local upward movements in Figure 4.4 require a distinction between physical degradation and measured available capacity. The plotted points originate from repeated capacity check-ups and are not model predictions. A later check-up can yield a temporarily higher capacity because relaxation and reversible recovery, conditioning before the test, small temperature differences, current-integration uncertainty, and voltage cut-off detection all affect the measured result.[17] The available data do not isolate one of these effects as the unique cause of every increase. The correct interpretation is therefore a downward long-term ageing trend with locally non-monotonic measurement anchors. Linear interpolation carries these anchor variations into the displayed SOH trajectories.

The robustness study uses a fixed cell-disjoint split consisting of six training cells, three validation cells, and six independent holdout test cells. Neural training, scaler fitting, pruning, model selection, and the identification of the HECM parameter surfaces use only the training and validation sets. The holdout cells are reserved for the final nominal, disturbance, recovery, and embedded comparisons. Table 4.3 documents the complete assignment and the measured duration, SOH, temperature,